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PROJECT REPORT ON

PSCS_35: 360-DEGREE FEEDBACK SOFTWARE

FOR NEWS STORIES IN REGIONAL MEDIA (USING AI/ML)

Department: Computer Science Engineering

Submitted by:

• **YASHAS RAJ** : 20231CSE3019 •

SACHIN KUMAR SAROJ : 20231CSE3018 •

RAKSHITHA REDDY KS : 20231CSE308

Abstract – Revolutionising Regional News Analysis

The project *NewsLensAI* presents an AI/ML-driven solution designed to analyze regional news stories and provide actionable insights for policymakers, media houses, and researchers. Traditional approaches to news review and sentiment analysis often rely on manual processes, which are time-consuming, subjective, and lack scalability. To address these challenges, this project leverages Natural Language Processing (NLP) techniques to perform automated sentiment classification (positive, neutral, negative) and department-wise categorization of news content (such as health, education, defense, etc.).

The system incorporates an interactive web-based dashboard that visualizes sentiment distribution and departmental mapping, enabling stakeholders to gain a comprehensive 360-degree view of public opinion on government policies and initiatives. By integrating machine learning models, TextBlob-based sentiment analysis, and keyword-driven classification, the project ensures faster, unbiased, and more reliable analysis of news stories.

Furthermore, the design emphasizes scalability and future enhancements, including multilingual support for regional Indian languages and real-time analysis capabilities. Overall, *NewsLensAI* aims to provide a cost-effective, user-friendly, and impactful tool for sentiment-driven decision-making, thereby improving the communication loop between governments, media, and the public.

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Sl. No.	Section	Description
1	Abstract	Overview of the project's purpose, scope, and significance.
2	Literature Survey (10 Papers)	Summary and insights from ten relevant IEEE research papers.
3	Objectives	Specific goals and outcomes expected from the project.
4	Existing Methodology and Drawbacks	Current approaches in use and their limitations.
5	Proposed Methods and Feasibility Study	The suggested solution and an evaluation of its practicality.
6	Architecture Diagram	Visual representation of the project workflow and system design.
7	Modules	Description of different modules and their functions in the system.
8	Hardware and Software Requirements	System and software specifications required for implementation.
9	Timeline (Gantt Chart)	Planned schedule of the project phases and deliverables.
10	References	List of research papers, articles, and other resources cited.

Literature Survey (

❑ Deciphering Political Entity Sentiment in News with LLMs (2024)

Objective:

This study focused on understanding sentiment at the entity level in news articles, specifically targeting political leaders, parties, and departments rather than analyzing the overall polarity of the entire document. The goal was to provide fine-grained insights into how specific actors are perceived in the media.

Methodology:

The authors leveraged large language models (LLMs) to identify political entities within news text. After extracting entities, sentiment classification was applied to each entity individually using annotated datasets. The evaluation compared entity-level sentiment detection with traditional document-level sentiment analysis to demonstrate the added granularity. They used fine-tuned LLMs for entity recognition and contextual sentiment understanding.

Findings:

Entity-level analysis provided more precise insights into political sentiment, revealing divergent opinions about different figures in the same news article. The study also showed that LLM-based approaches outperform traditional lexicon or ML-based methods in capturing nuanced opinions directed at specific entities.

Relevance

Incorporating entity-level tagging in NewsLensAI can help monitor department-specific sentiment or public reactions toward particular ministers and policies. This enables targeted insights for policymakers and better-informed feedback mechanisms. By applying entity-level sentiment, NewsLensAI can support nuanced reporting and policy analysis.

Prediction of Stock Value Changes Using Sentiment (2024)

Objective:

The paper examined the correlation between news sentiment and short-term stock market movements, exploring whether sentiment extracted from news and web data could predict stock value changes.

Methodology:

Researchers collected news articles and social media posts related to companies and extracted sentiment scores using NLP techniques. These sentiment features were integrated with statistical and machine learning models to forecast stock price trends. Model performance was validated against real-time stock market data, demonstrating predictive capabilities.

Findings:

The study found that positive or negative sentiment in news articles often aligns with short-term stock movements. ML models incorporating sentiment outperformed baseline models relying purely on historical price trends. Fine-grained sentiment features enhanced prediction accuracy, especially when combined with domain-specific financial indicators.

Relevance

The study highlights the practical impact of sentiment analysis beyond text interpretation. For NewsLensAI, integrating financial or policy-related sentiment can support decision-making dashboards, demonstrating how automated sentiment signals translate into measurable real-world effects.

Fake News Stance Detection Using Deep Learning (2023)

Objective:

This paper focused on detecting the stance of news content relative to specific claims, which can help identify misleading or false reporting. Stances were classified into support, oppose, or neutral categories.

Methodology:

The authors compiled datasets with claim-article pairs annotated for stance. Deep learning classifiers, including LSTM and transformer-based architectures, were trained to predict the stance of the news article relative to the claim. Model performance was benchmarked against traditional ML classifiers.

Findings:

Deep learning approaches significantly improved accuracy in identifying stance. Multi-class classification allowed for more nuanced detection of misleading or manipulative content. The method effectively distinguished supportive articles from those opposing a claim or remaining neutral.

Relevance

Incorporating stance detection enhances credibility assessment in NewsLensAI. It can complement sentiment analysis by flagging potentially misleading or biased content, ensuring users receive reliable and context-aware insights.

Deep Sentiment Analysis: Case Study on Hausa (2024)

Objective:

This study addressed sentiment analysis for Hausa, a low-resource language, highlighting preprocessing challenges and resource limitations in non-English languages.

Methodology:

Researchers applied targeted tokenization, stemming, and lexicon-based methods to build sentiment datasets. Manual annotation supplemented automated methods to create evaluation corpora. Deep learning models were then trained and evaluated on these curated datasets.

Findings:

The study demonstrated that careful preprocessing is crucial for low-resource languages. Incorporating linguistic features such as morphology and stemming improved sentiment classification accuracy. Despite resource constraints, effective sentiment models could be built with a combination of manual and automated methods.

Relevance

This work offers strategies for expanding NewsLensAI to regional languages in India. Adapting preprocessing pipelines and leveraging limited datasets ensures accurate sentiment detection even in languages with scarce computational resources.

Multilingual Sentiment Analysis (Survey, 2024)

Objective:

The survey reviewed existing approaches for multilingual sentiment analysis, comparing rule-based, ML, and deep learning methods. It also evaluated translation-based methods versus native language models.

Methodology:

The authors collected multilingual datasets across several languages and benchmarked various sentiment analysis techniques. They compared translation pipelines (e.g., translating text into English) against models trained directly on native language corpora. Evaluation included both accuracy and resource efficiency.

Findings:

Native models often outperformed translation-based approaches, particularly for nuanced sentiment. Rule-based systems were limited in scope, while deep learning approaches, especially transformer-based models, excelled in multilingual contexts. The trade-offs between accuracy and computational cost were highlighted.

Relevance

Guides the selection of models when expanding NewsLensAI beyond English. Accurate regional language sentiment coverage is essential for real-time monitoring of local news and regional policy reactions.

Enhancing Multilingual Hate Speech Detection (2024)

Objective:

The paper developed context-aware, multilingual hate speech detectors that account for cultural and linguistic nuances.

Methodology:

Researchers trained deep learning classifiers with multilingual embeddings on datasets from multiple languages. Contextual features were incorporated to distinguish harmful speech from neutral or benign expressions. The models were benchmarked against standard baselines for cross-lingual performance.

Findings:

Context-aware embeddings improved classification accuracy, especially in low-resource languages. Cross-lingual transfer enabled better performance for languages with limited training data.

Relevance Techniques for context-aware sentiment analysis and cross-lingual transfer can enhance NewsLensAI's ability to flag sensitive or harmful content in regional media, increasing robustness and reliability.

Sentiment Analysis of Tweets Using Emojis and Text (2024)

Objective:

The study explored how combining emoji usage with textual content improves sentiment classification in social media posts.

Methodology:

The authors built multimodal datasets that include both text and emojis. Classifiers incorporated emoji semantics alongside textual features. Performance was compared against text-only models on accuracy and F1 score metrics.

Findings:

Multimodal approaches outperformed text-only models, particularly for short, informal messages where emojis convey significant sentiment cues. The results highlight the importance of non-textual markers in sentiment analysis.

Relevance

Incorporating metadata and non-textual cues such as emojis, hashtags, or engagement metrics can improve sentiment prediction for short headlines, tweets, and social media excerpts relevant to news monitoring.

Financial Sentiment Analysis with LLMs & FinBERT (2024)

Objective:

The study compared general-purpose LLMs and domain-specific FinBERT models for fine-grained sentiment analysis on financial news.

Methodology:

FinBERT was fine-tuned on labeled financial corpora, while generic LLMs were applied without adaptation. The models were evaluated on datasets of news and financial statements for sentiment detection accuracy and granularity.

Findings:

Domain-specific FinBERT models outperformed general LLMs, especially in understanding financial jargon and subtle sentiment cues. Fine-tuning significantly enhanced performance on specialized datasets.

Relevance

Highlights the value of domain-adapted models. NewslensAI could benefit from fine-tuning sentiment models on governmental or administrative corpora for more accurate department-level sentiment insights.

Stock Prediction using Sentiment from Headlines (2024)

Objective:

This paper investigated the predictive power of short news headlines for downstream tasks such as stock value movement prediction.

Methodology:

Researchers collected headline datasets, extracted sentiment features, and applied deep learning models for prediction. The performance was compared with models using full-text news articles to test the efficiency of headline-based signals.

Findings:

Headlines alone were sufficient to provide strong sentiment signals for predictive tasks. The study validated that short snippets can contain concentrated sentiment information relevant for decision-making.

Relevance

Supports using headlines and short news excerpts in NewslensAI, which is particularly useful for monitoring regional news quickly and providing real-time insights.

Ensemble Learning for Multi-Modal Fake News Detection (2024)

Objective:

The study focused on combining multiple modalities (text, images, metadata) to improve fake news detection.

Methodology:

The authors built ensemble models integrating outputs from text, image, and metadata classifiers. Each modality was processed separately, and a weighted ensemble strategy was used to generate final predictions. The system was benchmarked against single-model baselines.

Findings:

Ensemble models outperformed individual classifiers, showing higher robustness against diverse misinformation formats. Multimodal signals provided complementary information, improving overall detection accuracy.

Relevance

Incorporating multimodal analysis could enhance NewsLensAI by combining text sentiment, images, and other metadata to assess news credibility and reduce the risk of false or misleading content dissemination.

Objectives – Project Goals

The primary goal of *NewsLensAI* is to develop an AI/ML-based system that can automatically analyze regional news articles and provide policymakers, media houses, and researchers with reliable, timely, and unbiased insights.

Specific Objectives

1. **Automated Sentiment Analysis**
 - Classify news content into positive, negative, and neutral sentiments using NLP and ML models.
2. **Departmental Classification**
 - Automatically categorize news stories into government-related departments (e.g., health, education, defense, infrastructure) for better policy alignment.
3. **Dashboard Development**
 - Build a user-friendly, web-based dashboard (using Flask and Chart.js) to visualize sentiment trends, departmental distributions, and overall public perception.
4. **Bias Reduction**
 - Eliminate manual review bias by automating the sentiment analysis process, ensuring consistency and objectivity.
5. **Scalability and Timeliness**
 - Provide real-time or near real-time insights to handle the high volume of regional news articles published daily.
6. **Foundation for Multilingual Support**
 - Design the system in a way that it can later be extended to cover multiple Indian regional languages.

Expected Outcomes

- Policymakers gain a **360-degree view** of public opinion on government schemes and policies.
- Media houses can access **unbiased insights** to enhance reporting quality.
- Researchers can study **sentiment trends** over time across departments and regions.

Existing Methodology and Drawbacks

Existing Approaches

1. Manual Review by Analysts

- News articles are manually read, interpreted, and categorized by human experts.
- Sentiment (positive/negative/neutral) is judged subjectively, often influenced by the analyst's perception.

2. Basic Sentiment Analysis Tools

- Some platforms use lexicon-based tools (e.g., simple word-counting of positive and negative terms).
- These tools lack contextual understanding and fail when news involves sarcasm, negation, or regional language nuances.

3. General-Purpose NLP Applications

- Off-the-shelf NLP models not designed for the government or regional media context are sometimes applied.
- They perform poorly in domain-specific classification (e.g., mapping articles to health, defense, or education departments).

Drawbacks

- **Time-Consuming & Expensive**

Manual analysis of daily news content requires large human teams, making the process slow and costly.

- **Human Bias**

Interpretations of sentiment can vary significantly across individuals, reducing reliability and objectivity.

- **Delayed Feedback**

By the time insights are compiled and shared, they may already be outdated, reducing their usefulness for real-time decision-making.

- **Limited Departmental Categorization**

Existing sentiment tools rarely classify content into departments or policy domains, limiting the scope of actionable insights.

- **Scalability Issues**

Current methodologies cannot efficiently handle the growing volume of digital regional news across multiple languages.

Proposed Methods and Feasibility Study

Proposed Methods

1. Data Collection

- News data will be collected from CSV files, web scraping (using *BeautifulSoup4*), or manual entry.
- Sources will primarily focus on regional media outlets to ensure relevance.

2. Preprocessing

- Cleaning text by removing noise (punctuation, stop words, irrelevant symbols).
- Tokenization and normalization to prepare data for analysis.
- Handling of special cases such as regional terms and context-specific keywords.

3. Sentiment Analysis

- Using *TextBlob* and ML models (e.g., logistic regression, Naive Bayes, or transformers for advanced stages).
- Classify articles into **positive, negative, or neutral** categories.

4. Departmental Classification

- Apply keyword-based rules (e.g., “hospital” → Health, “school” → Education).
- Use ML-based classifiers for more accurate department mapping.

5. Dashboard & Visualization

- Develop an interactive **Flask web app** integrated with *Chart.js*.
- Visualize sentiment distribution, departmental mapping, and trends over time.

6. Storage & Analytics

- Store processed data in **SQLite**, allowing persistence and historical trend analysis.
- Provide export options for policymakers and researchers.

Feasibility Study

• Technical Feasibility

- All tools (*Python, Flask, TextBlob, Pandas, Chart.js*) are open-source and well-documented, making them suitable for academic and production environments.
- Lightweight database (*SQLite*) ensures easy deployment without heavy infrastructure.

• Economic Feasibility

- The system runs on affordable hardware (Intel i3/Apple M1 with 8GB RAM).
- No additional licensing costs since all libraries are open-source.

Architecture Diagram

The architecture of *NewsLensAI* follows a **modular pipeline design**, ensuring scalability, maintainability, and clear workflow from raw data collection to actionable insights.

System Workflow Description

1. Data Collection Layer

- News articles are gathered from multiple sources including CSV datasets, web scraping (regional media portals), or manual entry.
- This ensures diverse and rich datasets covering various policy domains.

2. Preprocessing Layer

- Raw text is cleaned by removing stop words, punctuation, and irrelevant tokens.
- Tokenization, stemming, and normalization are applied to prepare the text for further analysis.
- Special handling is provided for regional terms and mixed-language data.

3. Sentiment Analysis Layer

- Natural Language Processing (NLP) models (e.g., *TextBlob*, machine learning classifiers) analyze the polarity of the news content.
- The output sentiment is classified into **Positive, Neutral, or Negative** categories.

4. Department Classification Layer

- A keyword-driven and ML-based model identifies the most relevant government department for each article.
- Example: News about hospitals → Health, schools → Education, defense deals → Defense.

5. Storage Layer

- Processed data and results are stored in an **SQLite database**.
- This ensures persistence, enabling historical analysis and longitudinal tracking of public sentiment.

6. Dashboard & Visualization Layer

- An interactive web dashboard is built using **Flask** and **Chart.js**.
- It presents sentiment distribution, departmental trends, and policy-specific public opinions in real time.

Key Advantages of the Architecture

- **Modularity** – Each layer is independent, making it easy to update or replace.
- **Scalability** – Can handle large volumes of regional news data.
- **Flexibility** – Supports integration of multilingual datasets and real-time streaming APIs in the future.
- **User-Centric** – The dashboard simplifies complex NLP analysis into intuitive visual insights for policymakers and researcher

Modules

The *NewsLensAI* system is organized into well-defined modules, each responsible for a specific function within the workflow. This modular design enhances clarity, flexibility, and maintainability.

1. Data Collection Module

- Collects news input from multiple sources: CSV datasets, web scraping, or manual entry.
- Ensures diverse data covering different government departments and regional policies.

2. Preprocessing Module

- Cleans the raw text by removing noise such as stop words, punctuation, and irrelevant tokens.
- Performs tokenization, stemming, and normalization for consistent representation.
- Handles multilingual and regional language complexities.

3. Sentiment Analysis Module

- Uses NLP and ML-based models (*TextBlob*, *Logistic Regression*, *Naive Bayes*, or advanced *LLMs*) to detect sentiment polarity.
- Classifies articles into **Positive**, **Negative**, or **Neutral** categories.
- Reduces manual bias by providing objective sentiment scoring.

4. Department Classification Module

- Applies keyword-based rules and ML-driven classifiers to map articles to government departments.
- Example mappings:
 - “Hospital, healthcare” → Health
 - “Schools, education policy” → Education
 - “Army, defense budget” → Defense
- Provides department-level categorization for policy evaluation.

5. Storage Module

- Stores processed data, sentiment scores, and department mappings in an **SQLite database**.
- Maintains historical records, enabling long-term trend analysis and comparative studies.

6. Dashboard & Visualization Module

- Built using **Flask** (backend) and **Chart.js** (frontend).
- Provides an interactive and user-friendly interface.
- Displays:
 - Sentiment distribution across articles.
 - Departmental classification results.
 - Time-based trends for policy evaluation.

Hardware and Software Requirements

The successful implementation of *NewsLensAI* depends on a combination of suitable hardware and software resources. The chosen specifications ensure cost-effectiveness, scalability, and ease of deployment.

Hardware Requirements

Component	Specification
Processor	Intel Core i3 / Apple M1 (or higher)
RAM	Minimum 8 GB (recommended 16 GB for larger datasets)
Storage	50 GB free disk space for datasets, preprocessing files, and database storage
Internet	Required for web scraping, API calls, and deployment
System	Laptop/Desktop (Mac Mini or equivalent can be used for lightweight deployment)

Software Requirements

Category	Specification
Operating System	macOS / Linux / Windows (cross-platform support)
Programming Lang.	Python 3.x
Frameworks	Flask (for web app development)
Libraries	TextBlob (sentiment analysis), Pandas (data handling), BeautifulSoup4 (scraping), Chart.js (visualization)
Database	SQLite (lightweight, built-in)
IDE/Editor	VS Code / PyCharm / Replit (for development and testing)

Justification

- **Cross-Platform Support** – Works across major operating systems (Windows, macOS, Linux).
- **Open-Source Tools** – All frameworks and libraries are open-source, ensuring no additional cost.
- **Lightweight Database** – SQLite ensures ease of integration without requiring heavy infrastructure

Timeline (Gantt Chart)

Phase	Activity	Tasks Included	Duration
1	Requirement Analysis & Planning	Define problem statement, finalize objectives, assign roles	Week 1 – 2
2	Literature Survey	Review 10 IEEE papers, identify gaps, document insights	Week 2 – 3
3	Data Collection	Gather datasets via CSV, scraping regional media portals, manual inputs	Week 3 – 4
4	Preprocessing	Text cleaning, tokenization, stop-word removal, stemming, normalization	Week 4 – 5
5	Model Development	Implement sentiment analysis & department classification (TextBlob/ML models)	Week 5 – 7
6	Dashboard Development	Flask web app design, Chart.js integration for visualization	Week 7 – 8
7	Database Integration	Store processed results in SQLite, enable historical trend analysis	Week 8 – 9
8	Testing & Validation	Accuracy evaluation, error handling, dashboard usability testing	Week 9 – 10
9	Documentation & Reporting	Prepare review reports, technical documentation, and module details	Week 10-11
10	Final Review & Deployment	Present project outcomes, finalize deployment, submit final report	Week 11 – 12

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